SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Computer Networks **COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO5: *Measure network performance using key metrics with simulation tools*

Assessment Tool Weightage: 20%

QUESTIONS: 40 **Total Marks:40**

Annexure A

Mock Interview

QUESTIONS:40 **Total Marks:40**

1. Analyze how network congestion affects packet delay and packet loss. (BL4)(1)
2. Compare the impact of star and mesh topologies on network reliability and fault tolerance. (BL4)(1)
3. Why does increasing the number of routers between source and destination increase end-to-end delay? Explain logically. (BL4)(1)
4. Analyze how Quality of Service (QoS) improves the performance of voice and video traffic. (BL4)(1)
5. Evaluate the impact of using fiber optic cables instead of copper cables for high-speed communication. (BL5)(1)
6. Analyze how network segmentation using VLANs improves performance and security. (BL4)(1)
7. Compare the effects of TCP and UDP on network performance for different applications. (BL4)(1)
8. Why is load balancing important in large enterprise networks? Analyze its impact on performance and availability. (BL5)(1)
9. Analyze how frequent routing updates influence network efficiency and bandwidth utilization. (BL4)(1)
10. Evaluate the trade-off between network security measures (such as firewalls and encryption) and overall network performance. (BL5)(1)
11. A network has high delay—how will you identify and fix the issue? (BL4)(1)
12. If throughput is low despite high bandwidth, what could be the reasons? (BL4)(1)
13. How would you reduce packet loss in a congested network? (BL4)(1)
14. A VoIP call is breaking frequently—how would you troubleshoot? (BL4)(1)
15. Given a network with high jitter, suggest improvements (BL4)(1)
16. How will you simulate a congested network in Packet Tracer? (BL5)(1)
17. If packets are delayed in a router, what steps will you take? (BL4)(1)
18. How would you optimize a network for online gaming? (BL4)(1)
19. Design a small network and explain how you will measure its performance (BL5)(1)
20. If RTT is very high, how will you diagnose the issue? (BL4)(1)
21. Explain network performance to a non-technical person (BL5)(1)
22. Describe throughput using a real-life example (BL5)(1)
23. Present how packet loss affects video streaming (BL5)(1)
24. Explain delay types clearly in under 1 minute (BL4)(1)
25. Describe Cisco Packet Tracer in a professional way (BL5)(1)
26. Explain QoS in simple and technical terms (BL5)(1)

27. How would you present your simulation results to a client? (BL4)(1)
28. Explain jitter using a real-world analogy (BL5)(1)
29. Describe how you tested network performance in a lab (BL4)(1)
30. Explain the importance of performance metrics in one structured answer (BL5)(1)
31. Introduce yourself and explain your knowledge of network performance (BL5)(1)
32. Why are you interested in networking and simulation tools? (BL5)(1)
33. What makes you confident in analyzing network performance? (BL5)(1)
34. Explain a concept you are very strong in (related to networking) (BL5)(1)
35. How would you handle a question you don't know? (BL5)(1)
36. Describe a situation where you solved a technical problem (BL4)(1)
37. Why should we select you for a networking role? (BL5)(1)
38. What challenges did you face while learning Packet Tracer? (BL5)(1)
39. How do you improve your technical skills regularly? (BL5)(1)
40. Demonstrate confidence while explaining a complex concept simply (BL5)(1)

RUBRICS FOR CALCULATION

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Conceptual Understanding	**30%**	**Demonstrates thorough understanding; answers all questions accurately**	**Shows good understanding with few errors**	**Partial understanding; several incorrect responses**	**Lacks understanding; majority of answers incorrect**
Accuracy of Answers	**25%**	**All answers are correct**	**Most answers are correct with minor mistakes**	**Some correct answers; frequent errors**	**Mostly incorrect answers**
Application of Knowledge	**20%**	**Correctly applies concepts to problem-based questions**	**Applies concepts with minor errors**	**Limited ability to apply concepts**	**Unable to apply concepts**
Time Management	**15%**	**Completes quiz well within time efficiently**	**Completes on time with minor delays**	**Requires extra time or rushes through questions**	**Unable to complete within given time**
Question Interpretation	**10%**	**Interprets all questions correctly and responds appropriately**	**Misinterprets a few questions**	**Often misinterprets questions**	**Unable to understand questions**

1. How does network congestion affect packet delay and packet loss?
Answer:
Network congestion occurs when too much traffic is present in the network. Routers become overloaded and packets wait in queues, which increases delay. If the buffers become full, packets are dropped, causing packet loss and retransmissions.

2. Compare star and mesh topologies in terms of reliability and fault tolerance.
Answer:
A star topology depends on a central switch, so if the switch fails, the entire network can be affected. However, failure of one individual cable affects only that device. A mesh topology has multiple paths between devices, so it provides higher reliability and fault tolerance, but it is more expensive and complex.

3. Why does increasing the number of routers increase end-to-end delay?
Answer:
Every router introduces processing and queuing delay. Packets must be received, examined, and forwarded at each router. Therefore, as the number of routers increases, the total end-to-end delay generally increases.

4. How does QoS improve voice and video traffic?
Answer:
QoS, or Quality of Service, prioritizes important traffic such as voice and video. It can give these packets higher priority and reduce delay, jitter, and packet loss. This results in clearer VoIP calls and smoother video streaming.

5. Evaluate fiber optic cables compared with copper cables for high-speed communication.
Answer:
Fiber optic cables provide much higher bandwidth and support longer distances with low signal loss. They are also less affected by electromagnetic interference. Copper is cheaper and easier to install, but fiber is generally better for high-speed and long-distance communication.

6. How does VLAN-based network segmentation improve performance and security?
Answer:
VLANs divide a network into logical segments. This reduces unnecessary broadcast traffic and improves performance. From a security perspective, users and devices can be separated into different VLANs, limiting unwanted access between groups.

7. Compare TCP and UDP for different applications.
Answer:
TCP is connection-oriented and provides reliable and ordered delivery, so it is suitable for web browsing, file transfer, and email. UDP is faster and has lower overhead but does not guarantee delivery, making it useful for streaming, VoIP, DNS, and online gaming.

8. Why is load balancing important in large enterprise networks?
Answer:
Load balancing distributes network traffic across multiple servers or network paths. It prevents one resource from becoming overloaded, improves response time, and increases availability. If one server fails, traffic can be redirected to another available server.

9. How do frequent routing updates influence network efficiency?
Answer:
Routing updates help routers maintain accurate information about network paths. However, if updates occur too frequently, they consume bandwidth and router processing resources. Therefore, routing protocols need to balance routing accuracy with control traffic overhead.

10. Evaluate the trade-off between security and network performance.
Answer:
Security measures such as firewalls, encryption, and intrusion detection improve protection but require additional processing. This can introduce some latency and reduce throughput. However, the security benefits are usually essential, so the goal is to implement efficient security without unnecessarily affecting performance.

Troubleshooting Questions
11. A network has high delay. How will you identify and fix the issue?
Answer:
First, I would use tools such as ping and traceroute to identify where the delay occurs. Then I would check bandwidth utilization, router CPU usage, congestion, and routing paths. After identifying the cause, I could optimize routing, remove bottlenecks, increase bandwidth, or apply QoS.

12. If throughput is low despite high bandwidth, what could be the reasons?
Answer:
High bandwidth does not always mean high throughput. Possible reasons include congestion, packet loss, high latency, inefficient protocols, hardware limitations, interference, or server-side limitations. I would measure actual traffic and identify the bottleneck.

13. How would you reduce packet loss in a congested network?
Answer:
I would first identify the source of congestion. Then I could increase bandwidth, apply QoS, optimize routing, use load balancing, or reduce unnecessary traffic. I would also check for faulty cables, interfaces, and overloaded network devices.

14. A VoIP call is breaking frequently. How would you troubleshoot it?
Answer:
I would check packet loss, latency, and jitter using network monitoring tools. Then I would check bandwidth utilization, Wi-Fi or cable quality, and router configuration. I would also prioritize VoIP traffic using QoS.

15. Given a network with high jitter, suggest improvements.
Answer:
I would identify congestion and inconsistent traffic flow first. Then I would apply QoS to prioritize real-time traffic, reduce congestion, optimize routing, and ensure sufficient bandwidth. For wireless networks, I would also check interference and signal quality.

16. How will you simulate a congested network in Packet Tracer?
Answer:
I would create multiple devices and generate traffic through a limited-bandwidth link. By increasing the traffic load, I can create congestion and observe packet delay or packet loss. I can then modify bandwidth or QoS settings and compare the results.

17. If packets are delayed in a router, what steps will you take?
Answer:

I would check the router's CPU and memory utilization, interface status, queue length, and traffic volume. I would also check the routing table and interface errors. If congestion is the cause, I would optimize routing, apply QoS, or increase capacity.

18. How would you optimize a network for online gaming?
Answer:
For gaming, low latency and low jitter are more important than extremely high bandwidth. I would use a reliable wired connection, prioritize gaming traffic using QoS, reduce background downloads, select efficient routes, and minimize congestion.

19. Design a small network and explain how you will measure its performance.
Answer:
I would create a network with PCs, a switch, a router, and a server. I would measure latency using ping, packet loss using packet statistics, throughput using data transfer measurements, and jitter for real-time traffic. I would compare the results under normal and high-traffic conditions.

20. If RTT is very high, how will you diagnose the issue?
Answer:
I would use ping and traceroute to identify where the delay occurs. Then I would check each hop for congestion, routing problems, high utilization, or long physical distance. After finding the bottleneck, I would optimize the route or reduce congestion.

21. Explain network performance to a non-technical person.
Answer:
I would explain network performance as how well a network delivers information. For example, a good network is like a wide, clear highway where vehicles move quickly without traffic jams. A poor network is like a crowded road where information takes longer to reach its destination.

22. Describe throughput using a real-life example.
Answer:
Throughput is the actual amount of data successfully transferred in a given period. For example, if a water pipe can theoretically carry 100 liters per minute but only 70 liters actually reach the destination, the effective flow is like network throughput.

23. Explain how packet loss affects video streaming.
Answer:
Video is divided into packets before transmission. If some packets are lost, the receiver may not get all the information required to display the video correctly. This can cause buffering, freezing, pixelation, or reduced video quality.

24. Explain delay types clearly in under one minute.
Answer:
There are mainly four types of network delay: processing, queuing, transmission, and propagation delay. Processing delay occurs when a device processes a packet. Queuing delay occurs when a packet waits in a queue. Transmission delay is the time needed to place the packet onto the link. Propagation delay is the time required for the signal to travel through the medium.

25. Describe Cisco Packet Tracer professionally.

Answer:
Cisco Packet Tracer is a network simulation and learning tool developed by Cisco. It allows us to design network topologies, configure routers and switches, simulate packet transmission, and troubleshoot network problems without requiring physical networking equipment.

26. Explain QoS in simple and technical terms.
Answer:
Simply, QoS means giving priority to important network traffic. Technically, Quality of Service uses mechanisms such as classification, prioritization, queuing, and traffic shaping to manage network resources and provide better performance for delay-sensitive applications such as VoIP and video.

27. How would you present your simulation results to a client?
Answer:
I would present the results using simple graphs and key performance metrics such as latency, throughput, packet loss, and jitter. I would compare the results before and after optimization and clearly explain the improvements and remaining issues in business-friendly language.

28. Explain jitter using a real-world analogy.
Answer:
Jitter is the variation in packet arrival time. Imagine a person receiving 10 passengers for a meeting. If everyone arrives exactly every 5 minutes, there is no jitter. If some arrive after 2 minutes, others after 10 minutes, the arrival time is inconsistent. That inconsistency is similar to network jitter.

29. Describe how you tested network performance in a lab.
Answer:
I created the network topology in Packet Tracer and configured the required IP addresses and routing. Then I tested connectivity using ping and observed packet transmission. I measured delay, packet loss, and traffic behavior under different network conditions and analyzed the results.

30. Explain the importance of performance metrics in one structured answer.
Answer:
Network performance metrics help us understand how efficiently a network operates. Latency measures delay, throughput measures successful data transfer, packet loss measures packets that fail to reach the destination, and jitter measures variation in packet arrival time. These metrics help identify problems and improve network quality.

31. Introduce yourself and explain your knowledge of network performance.
Answer:
Good morning. My name is Subahan. I am a computer science student with an interest in networking and cloud technologies. I have learned concepts such as network performance, TCP/IP, routing, switching, QoS, and network topologies. I have also worked with Cisco Packet Tracer to design and simulate networks. I am interested in improving my practical networking and troubleshooting skills.

32. Why are you interested in networking and simulation tools?
Answer:
I am interested in networking because it is the foundation that allows different systems and devices to communicate. Simulation tools such as Cisco Packet Tracer allow me to practice configurations and troubleshoot problems without needing physical hardware. I enjoy learning by creating and testing networks practically.

33. What makes you confident in analyzing network performance?
Answer:
My confidence comes from understanding the basic performance metrics and applying them practically. I can analyze latency, throughput, packet loss, and jitter and use tools such as ping, traceroute, and Packet Tracer to identify problems. I also try to approach problems logically instead of making assumptions.

34. Explain a concept you are very strong in related to networking.
Answer:
One concept I am comfortable with is QoS. QoS helps manage network traffic by giving priority to important applications. For example, voice and video require low delay and jitter, so QoS can prioritize them over less time-sensitive traffic. This helps maintain good application performance during congestion.

35. How would you handle a question you don't know?
Answer:
I would be honest rather than guessing. I would say, "I am not completely sure about that, but based on my understanding..." and explain what I know. I would then try to learn the correct answer. I believe admitting a knowledge gap and learning from it is better than providing incorrect information.

36. Describe a situation where you solved a technical problem.
Answer:
During a networking lab, if a device could not communicate with another device, I would troubleshoot it systematically. I would first check the physical or simulated connection, then IP addresses, subnet masks, gateway configuration, and routing. By checking each layer step by step, I could identify the configuration error and restore connectivity.

37. Why should we select you for a networking role?
Answer:
You should select me because I have a strong interest in networking, a good understanding of fundamental concepts, and a willingness to learn. I can approach technical problems logically and I am comfortable practicing with simulation tools. I may still be developing my experience, but I am committed to improving and contributing to the team.

38. What challenges did you face while learning Packet Tracer?
Answer:
Initially, I found router and switch configuration commands and troubleshooting difficult. I overcame this by practicing small networks first and gradually increasing their complexity. I also learned to troubleshoot systematically instead of changing multiple configurations at once.

39. How do you improve your technical skills regularly?
Answer:
I improve my skills through regular practice, online learning, lab simulations, and solving technical problems. Instead of only studying theory, I try to implement concepts using tools such as Packet Tracer. I also review my mistakes because they help me understand concepts better.

40. Demonstrate confidence while explaining a complex concept simply.
Answer:
One example is network congestion. I would explain it like this: imagine a highway with more vehicles than it can handle. Traffic slows down, vehicles wait longer, and some may not reach their destination efficiently. Similarly, when too much data enters a network, packets experience higher delay and some packets

may be dropped. We can solve this using techniques such as QoS, traffic management, and increased network capacity.